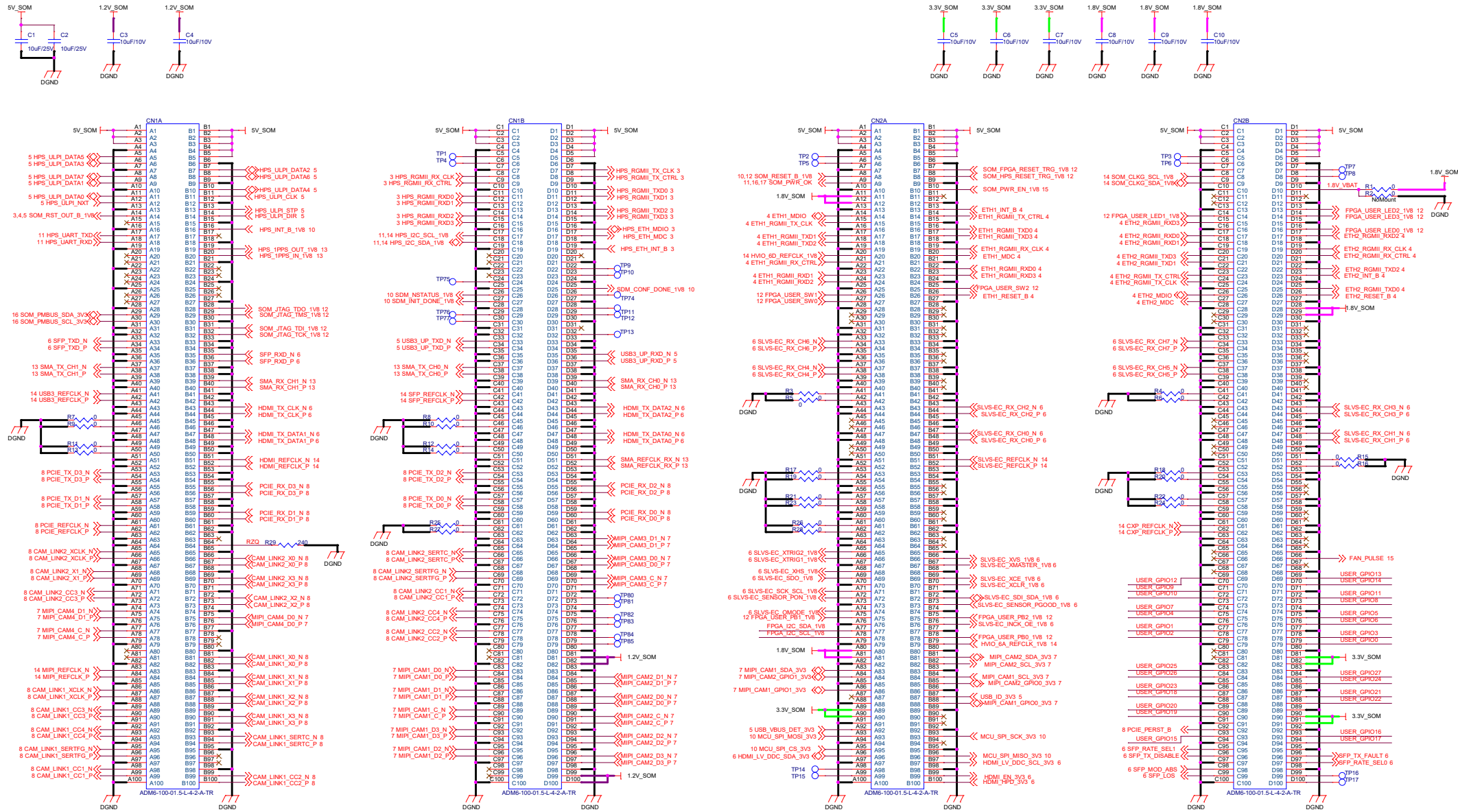
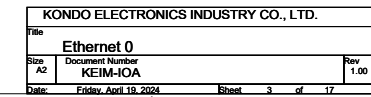
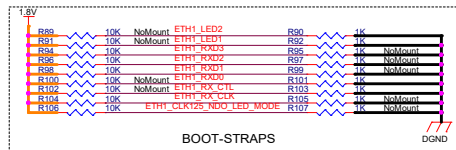




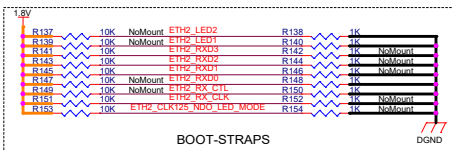
BtoB Connector #1

BtoB Connector #2

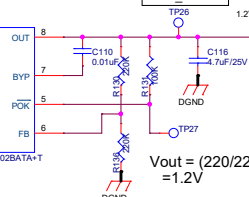
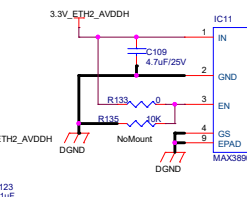
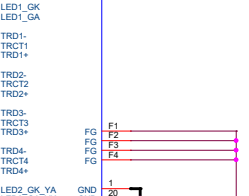
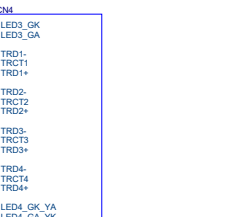
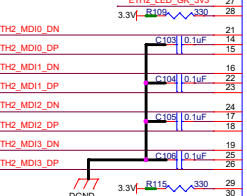
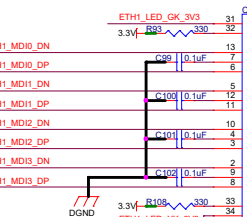
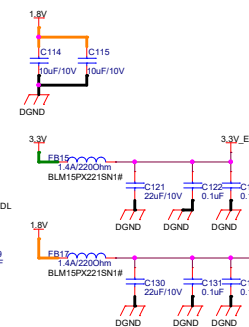
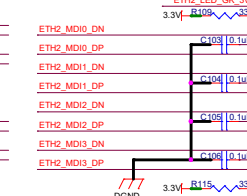
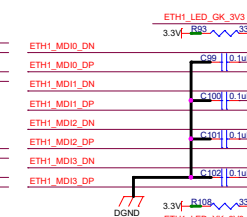
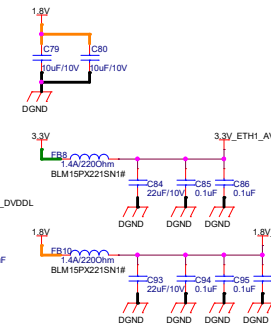
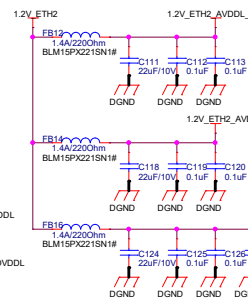
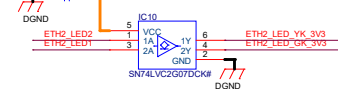
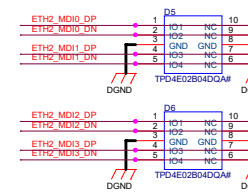
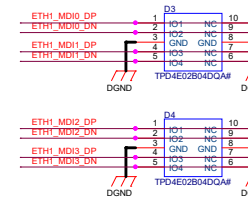
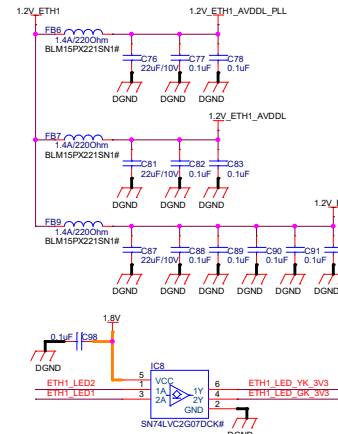


[illegible]

PHY ADD	0x04
MODE	RGMII mode advertise all capabilities (10/100 Full & Half, 1000 BT Full)
CLK125_EN	Disable 125MHz clock output
LED MODE	Single LED mode

[illegible]

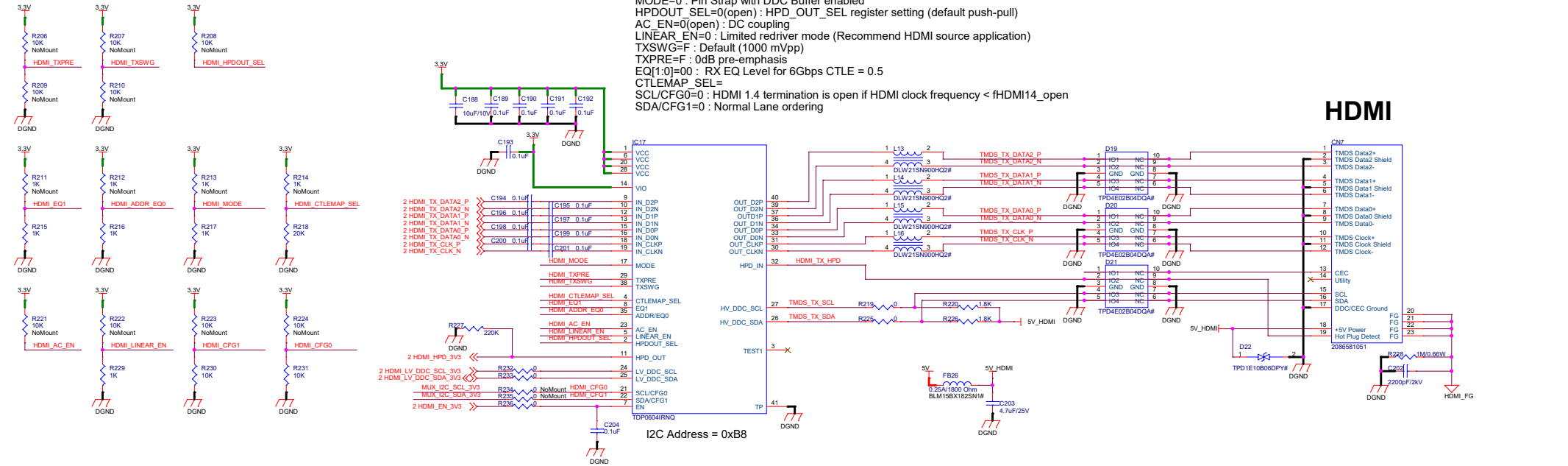
PHY ADD	0x04
MODE	RGMIi mode advertise all capabilities (10/100 Full & Half, 1000 BT Full)
CLK125_EN	Disable 125MHz clock output
LED MODE	Single LED mode



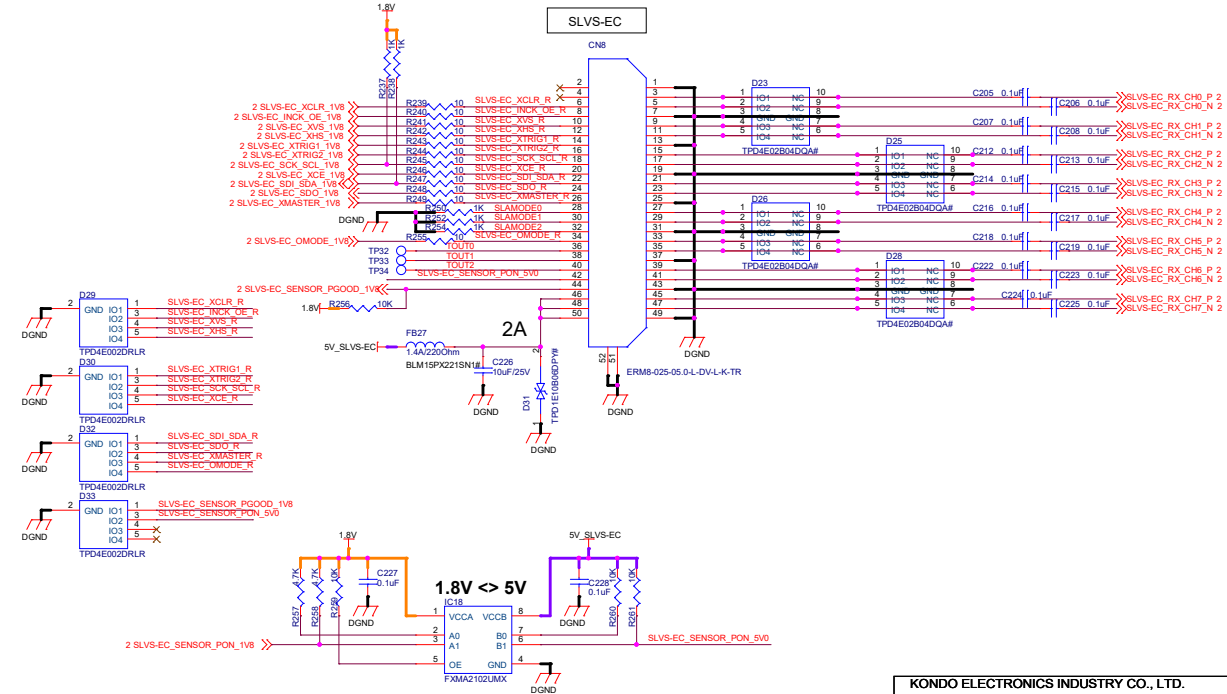
Pin-Strap Mode

MODE=0 : Pin Strap with DDC Buffer enabled
HPDOUT_SEL=0(open) : HPD_OUT_SEL register setting (default push-pull)
AC_EN=0(open) : DC coupling
LINEAR_EN=0 : Limited redriver mode (Recommend HDMI source application)
TXSWG=F : Default (1000 mVpp)
TXPRE=F : 0dB pre-emphasis
EQ[1:0]=00 : RX EQ Level for 6Gbps CTLE = 0.5
CTLEMAP_SEL=
SCL/CFG0=0 : HDMI 1.4 termination is open if HDMI clock frequency < fHDMI14_open
SDA/CFG1=0 : Normal Lane ordering

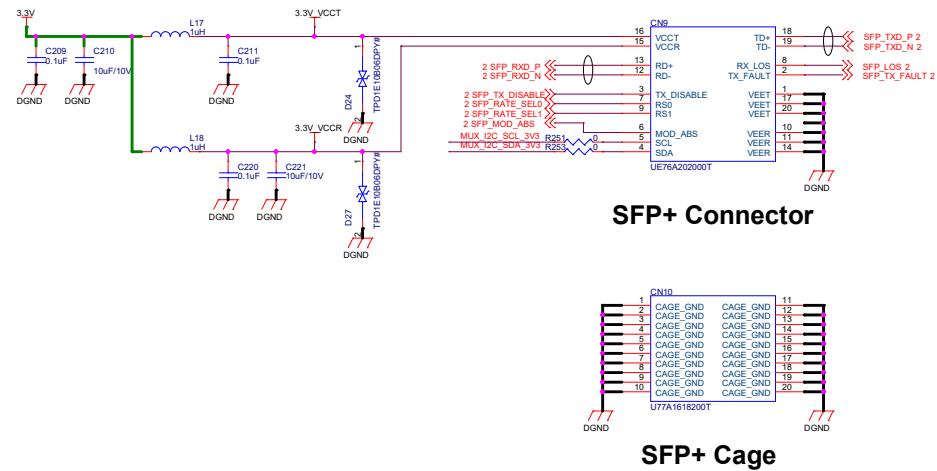
HDMI



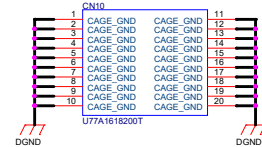
SLVS-EC



10GE

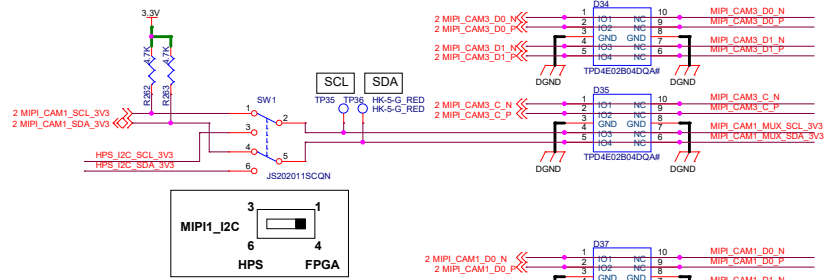


SFP+ Connector



SFP+ Cage

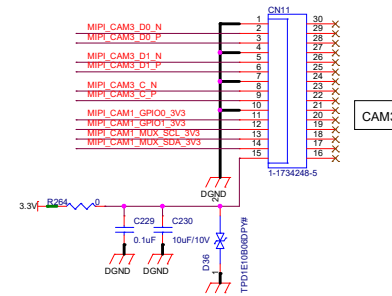
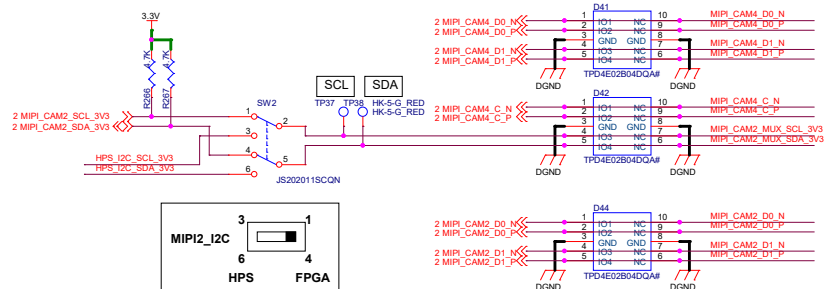




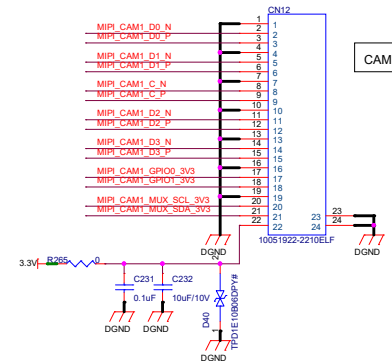
Shares GPIO and I2C between MIPI_CAM1 and MIPI_CAM3.

Shares GPIO and I2C between MIPI_CAM2 and MIPI_CAM4.

2 HPS_I2C_SCL_3V3
2 HPS_I2C_SDA_3V3



CAM3



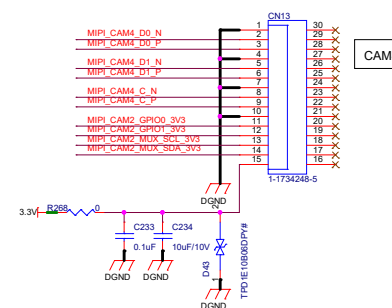
CAM1

MIPI15 connector pinout

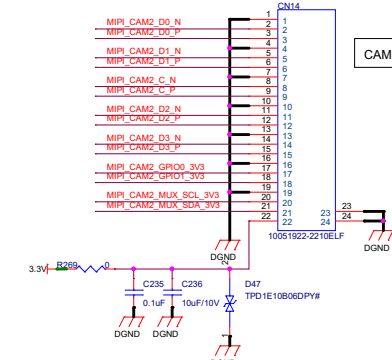
- 1: GND
- 2: CAM_D0_N
- 3: CAM_D0_P
- 4: GND
- 5: CAM_D1_N
- 6: CAM_D1_P
- 7: GND
- 8: CAM_CK_N
- 9: CAM_CK_P
- 10: GND
- 11: CAM_IO0
- 12: CAM_IO1
- 13: CAM_SCL
- 14: CAM_SDA
- 15: CAM_3V3

MIPI22 connector pinout

- 1: GND
- 2: CAM_D0_N
- 3: CAM_D0_P
- 4: GND
- 5: CAM_D1_N
- 6: CAM_D1_P
- 7: GND
- 8: CAM_CK_N
- 9: CAM_CK_P
- 10: GND
- 11: CAM_D2_N
- 12: CAM_D2_P
- 13: GND
- 14: CAM_D3_N
- 15: CAM_D3_P
- 16: GND
- 17: CAM_IO0
- 18: CAM_IO1
- 19: GND
- 20: CAM_SCL
- 21: CAM_SDA
- 22: CAM_3V3



CAM4



CAM2

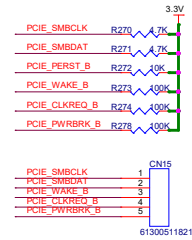
The diagram illustrates the pinout for a 16-lane PCIe card edge connector. It shows the connection of various signals to the card's pins, organized into groups for power, management, and data lanes.

Power and Management Signals:

- 3.3V_PCIE:** Connected to pins B1, B2, B3, B4, B5, B6, B7, B8, B9, B10, B11, B12, B13, B14, B15, B16, B17, B18, B19, B20, B21, B22, B23, B24, B25, B26, B27, B28, B29, B30, B31, B32, B33, B34, B35, B36, B37, B38, B39, B40, B41, B42, B43, B44, B45, B46, B47, B48, B49, B50, B51, B52, B53, B54, B55, B56, B57, B58, B59, B60, B61, B62, B63, B64, B65, B66, B67, B68, B69, B70, B71, B72, B73, B74, B75, B76, B77, B78, B79, B80, B81, B82, B83, B84, B85, B86, B87, B88, B89, B90, B91, B92, B93, B94, B95, B96, B97, B98, B99, B100, B101, B102, B103, B104, B105, B106, B107, B108, B109, B110, B111, B112, B113, B114, B115, B116, B117, B118, B119, B120, B121, B122, B123, B124, B125, B126, B127, B128, B129, B130, B131, B132, B133, B134, B135, B136, B137, B138, B139, B140, B141, B142, B143, B144, B145, B146, B147, B148, B149, B150, B151, B152, B153, B154, B155, B156, B157, B158, B159, B160, B161, B162, B163, B164, B165, B166, B167, B168, B169, B170, B171, B172, B173, B174, B175, B176, B177, B178, B179, B180, B181, B182, B183, B184, B185, B186, B187, B188, B189, B190, B191, B192.
- 12V_PCIE:** Connected to pins A1, A2, A3, A4, A5, A6, A7, A8, A9, A10, A11, A12, A13, A14, A15, A16, A17, A18, A19, A20, A21, A22, A23, A24, A25, A26, A27, A28, A29, A30, A31, A32, A33, A34, A35, A36, A37, A38, A39, A40, A41, A42, A43, A44, A45, A46, A47, A48, A49, A50, A51, A52, A53, A54, A55, A56, A57, A58, A59, A60, A61, A62, A63, A64, A65, A66, A67, A68, A69, A70, A71, A72, A73, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, A84, A85, A86, A87, A88, A89, A90, A91, A92, A93, A94, A95, A96, A97, A98, A99, A100, A101, A102, A103, A104, A105, A106, A107, A108, A109, A110, A111, A112, A113, A114, A115, A116, A117, A118, A119, A120, A121, A122, A123, A124, A125, A126, A127, A128, A129, A130, A131, A132, A133, A134, A135, A136, A137, A138, A139, A140, A141, A142, A143, A144, A145, A146, A147, A148, A149, A150, A151, A152, A153, A154, A155, A156, A157, A158, A159, A160, A161, A162, A163, A164, A165, A166, A167, A168, A169, A170, A171, A172, A173, A174, A175, A176, A177, A178, A179, A180, A181, A182, A183, A184, A185, A186, A187, A188, A189, A190, A191, A192.
- DGND:** Connected to pins B1, B2, B3, B4, B5, B6, B7, B8, B9, B10, B11, B12, B13, B14, B15, B16, B17, B18, B19, B20, B21, B22, B23, B24, B25, B26, B27, B28, B29, B30, B31, B32, B33, B34, B35, B36, B37, B38, B39, B40, B41, B42, B43, B44, B45, B46, B47, B48, B49, B50, B51, B52, B53, B54, B55, B56, B57, B58, B59, B60, B61, B62, B63, B64, B65, B66, B67, B68, B69, B70, B71, B72, B73, B74, B75, B76, B77, B78, B79, B80, B81, B82, B83, B84, B85, B86, B87, B88, B89, B90, B91, B92, B93, B94, B95, B96, B97, B98, B99, B100, B101, B102, B103, B104, B105, B106, B107, B108, B109, B110, B111, B112, B113, B114, B115, B116, B117, B118, B119, B120, B121, B122, B123, B124, B125, B126, B127, B128, B129, B130, B131, B132, B133, B134, B135, B136, B137, B138, B139, B140, B141, B142, B143, B144, B145, B146, B147, B148, B149, B150, B151, B152, B153, B154, B155, B156, B157, B158, B159, B160, B161, B162, B163, B164, B165, B166, B167, B168, B169, B170, B171, B172, B173, B174, B175, B176, B177, B178, B179, B180, B181, B182, B183, B184, B185, B186, B187, B188, B189, B190, B191, B192.

PCIe Signals:

- PCIE_SMBCLK:** Connected to pins B1, B2, B3, B4, B5, B6, B7, B8, B9, B10, B11, B12, B13, B14, B15, B16, B17, B18, B19, B20, B21, B22, B23, B24, B25, B26, B27, B28, B29, B30, B31, B32, B33, B34, B35, B36, B37, B38, B39, B40, B41, B42, B43, B44, B45, B46, B47, B48, B49, B50, B51, B52, B53, B54, B55, B56, B57, B58, B59, B60, B61, B62, B63, B64, B65, B66, B67, B68, B69, B70, B71, B72, B73, B74, B75, B76, B77, B78, B79, B80, B81, B82, B83, B84, B85, B86, B87, B88, B89, B90, B91, B92, B93, B94, B95, B96, B97, B98, B99, B100, B101, B102, B103, B104, B105, B106, B107, B108, B109, B110, B111, B112, B113, B114, B115, B116, B117, B118, B119, B120, B121, B122, B123, B124, B125, B126, B127, B128, B129, B130, B131, B132, B133, B134, B135, B136, B137, B138, B139, B140, B141, B142, B143, B144, B145, B146, B147, B148, B149, B150, B151, B152, B153, B154, B155, B156, B157, B158, B159, B160, B161, B162, B163, B164, B165, B166, B167, B168, B169, B170, B171, B172, B173, B174, B175, B176, B177, B178, B179, B180, B181, B182, B183, B184, B185, B186, B187, B188, B189, B190, B191, B192.
- PCIE_SMBDATA:** Connected to pins B1, B2, B3, B4, B5, B6, B7, B8, B9, B10, B11, B12, B13, B14, B15, B16, B17, B18, B19, B20, B21, B22, B23, B24, B25, B26, B27, B28, B29, B30, B31, B32, B33, B34, B35, B36, B37, B38, B39, B40, B41, B42, B43, B44, B45, B46, B47, B48, B49, B50, B51, B52, B53, B54, B55, B56, B57, B58, B59, B60, B61, B62, B63, B64, B65, B66, B67, B68, B69, B70, B71, B72, B73, B74, B75, B76, B77, B78, B79, B80, B81, B82, B83, B84, B85, B86, B87, B88, B89, B90, B91, B92, B93, B94, B95, B96, B97, B98, B99, B100, B101, B102, B103, B1



SW3 SETTINGS:
DEFAULT:1-OFF,2-ON

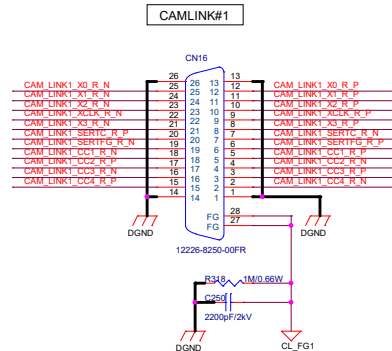
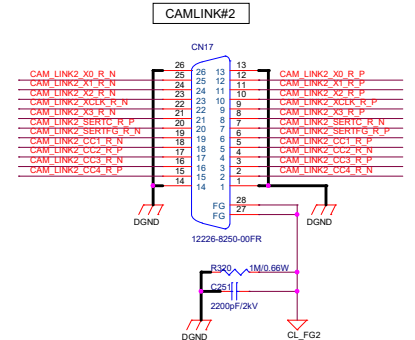


Figure 10 shows six pin connection diagrams for the TP4E02B04DQAI component, labeled D49, D51, D53, D55, D57, and D59. Each diagram shows a component with pins 1 through 10 connected to various signals and ground (GND).

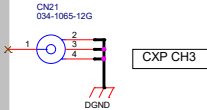
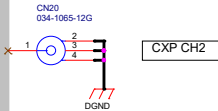
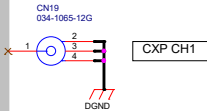
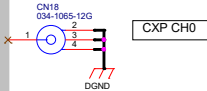
- D49:**
 - Pin 1: 2 CAM_LINK2_X0_N, 2 CAM_LINK2_X0_P
 - Pin 2: R282, R283
 - Pin 3: 1 I01 NC, 1 I02 NC, 1 I03 NC, 1 I04 NC
 - Pin 4: GND
 - Pin 5: GND
 - Pin 6: GND
 - Pin 7: GND
 - Pin 8: CAM_LINK2_X1_R_N, CAM_LINK2_X1_R_P
 - Pin 9: GND
 - Pin 10: GND
- D51:**
 - Pin 1: 2 CAM_LINK2_X2_N, 2 CAM_LINK2_X2_P
 - Pin 2: R290, R291
 - Pin 3: 1 I01 NC, 1 I02 NC, 1 I03 NC, 1 I04 NC
 - Pin 4: GND
 - Pin 5: GND
 - Pin 6: GND
 - Pin 7: GND
 - Pin 8: CAM_LINK2_XCLK_R_N, CAM_LINK2_XCLK_R_P
 - Pin 9: GND
 - Pin 10: GND
- D53:**
 - Pin 1: 2 CAM_LINK2_X3_N, 2 CAM_LINK2_X3_P
 - Pin 2: R298, R299
 - Pin 3: 1 I01 NC, 1 I02 NC, 1 I03 NC, 1 I04 NC
 - Pin 4: GND
 - Pin 5: GND
 - Pin 6: GND
 - Pin 7: GND
 - Pin 8: CAM_LINK2_SERTC_R_N, CAM_LINK2_SERTC_R_P
 - Pin 9: GND
 - Pin 10: GND
- D55:**
 - Pin 1: 2 CAM_LINK2_SERTCF_G_R_N, 2 CAM_LINK2_SERTCF_G_R_P
 - Pin 2: R306, R307
 - Pin 3: 1 I01 NC, 1 I02 NC, 1 I03 NC, 1 I04 NC
 - Pin 4: GND
 - Pin 5: GND
 - Pin 6: GND
 - Pin 7: GND
 - Pin 8: CAM_LINK2_CC1_R_N, CAM_LINK2_CC1_R_P
 - Pin 9: GND
 - Pin 10: GND
- D57:**
 - Pin 1: 2 CAM_LINK2_CC2_R_N, 2 CAM_LINK2_CC2_R_P
 - Pin 2: R314, R315
 - Pin 3: 1 I01 NC, 1 I02 NC, 1 I03 NC, 1 I04 NC
 - Pin 4: GND
 - Pin 5: GND
 - Pin 6: GND
 - Pin 7: GND
 - Pin 8: CAM_LINK2_CC3_R_N, CAM_LINK2_CC3_R_P
 - Pin 9: GND
 - Pin 10: GND
- D59:**
 - Pin 1: 2 CAM_LINK2_CC4_R_N, 2 CAM_LINK2_CC4_R_P
 - Pin 2: R324, R325
 - Pin 3: 1 I01 NC, 1 I02 NC, 1 I03 NC, 1 I04 NC
 - Pin 4: GND
 - Pin 5: GND
 - Pin 6: GND
 - Pin 7: GND
 - Pin 8: CAM_LINK2_CC4_R_N, CAM_LINK2_CC4_R_P
 - Pin 9: GND
 - Pin 10: GND

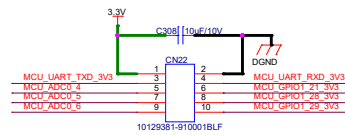




Confidential

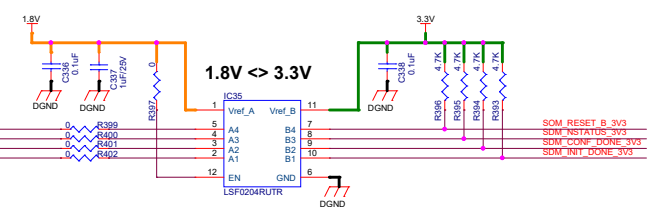
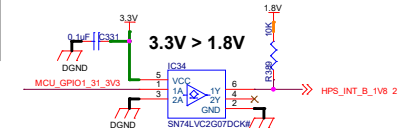
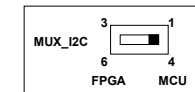
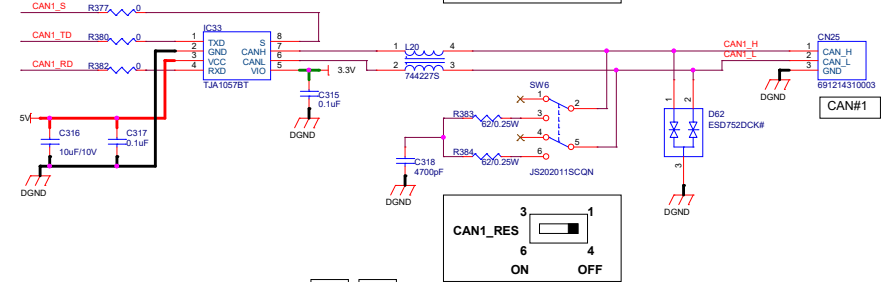
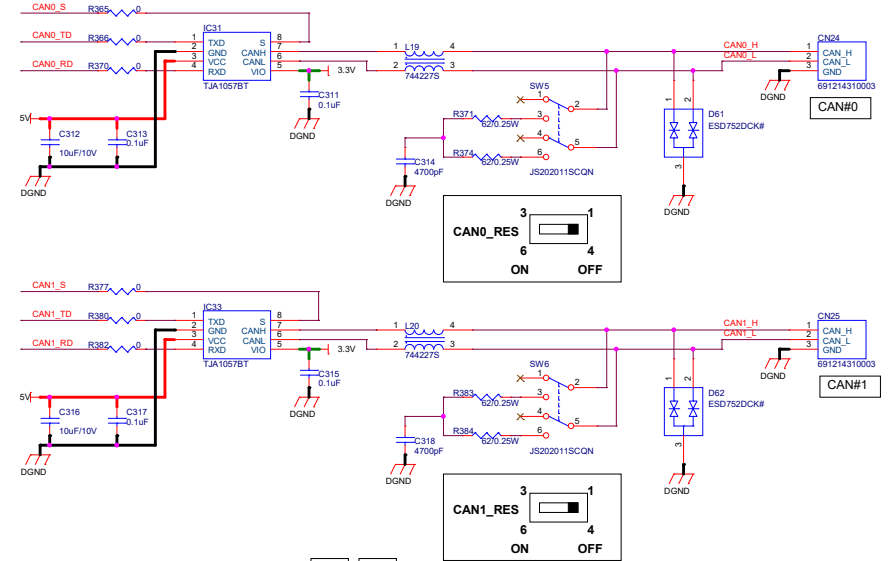
Contact us

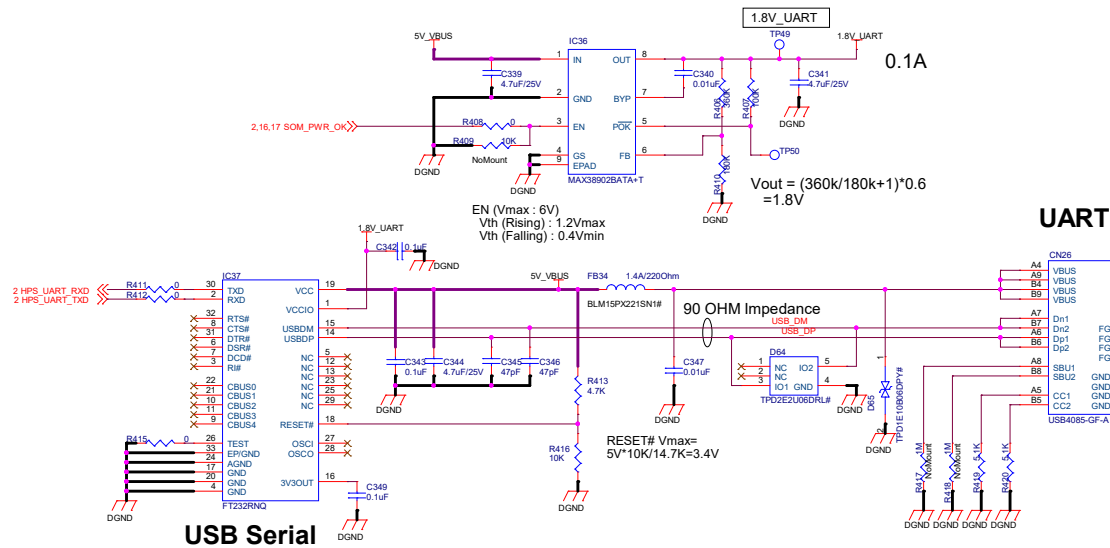




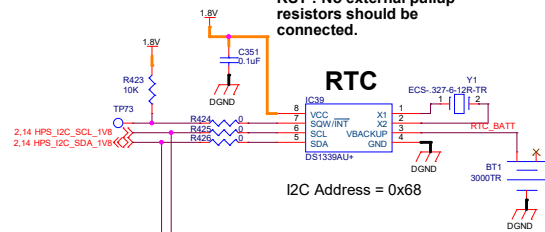
Enable ISP mode

1-High: Boot From internal Flash
1-Low: Enable ISP mode (UART)
2-Default High (Don't care.)

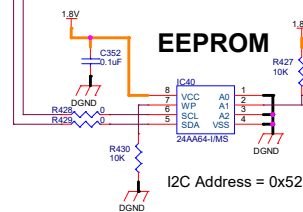




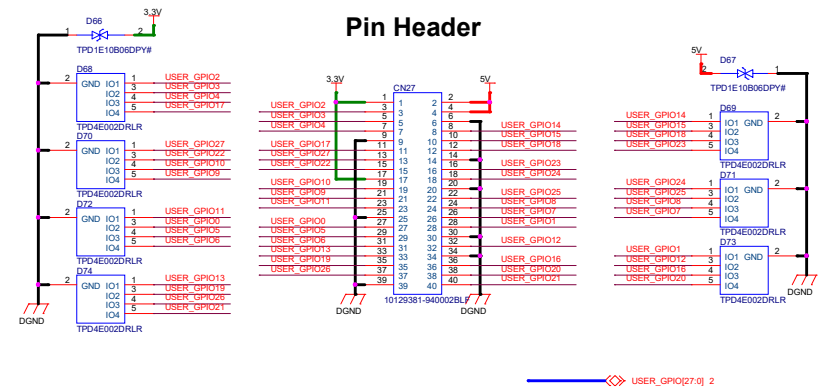
RST : No external pullup resistors should be connected.



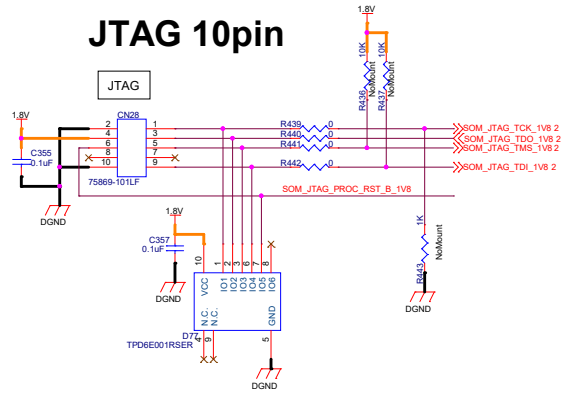
EEPROM



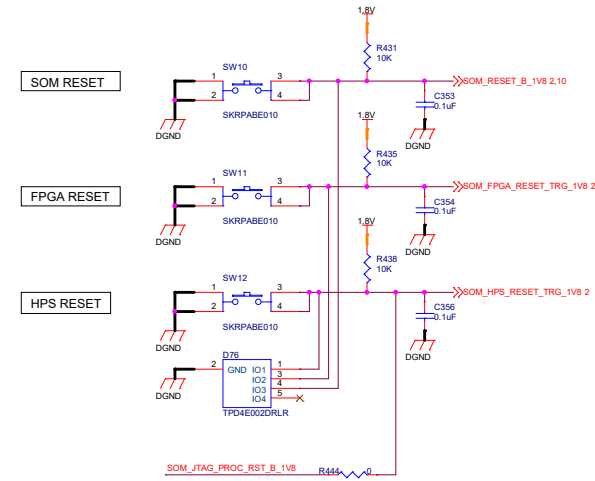
Pin Header



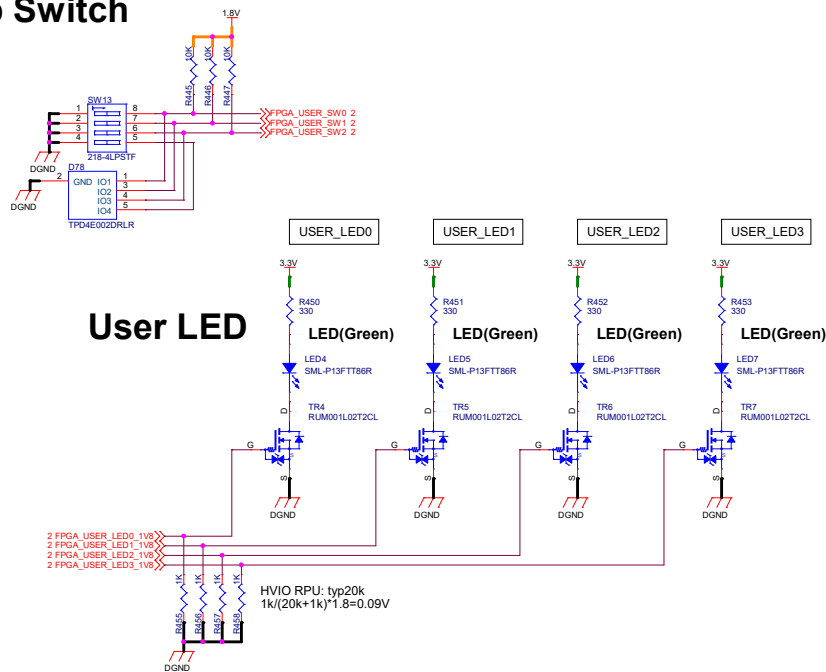
JTAG 10pin



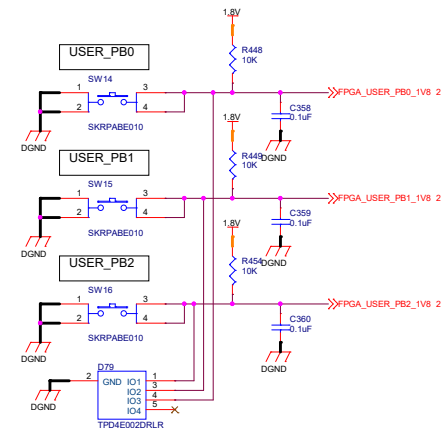
Reset Switch



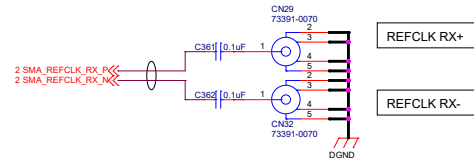
User Dip Switch



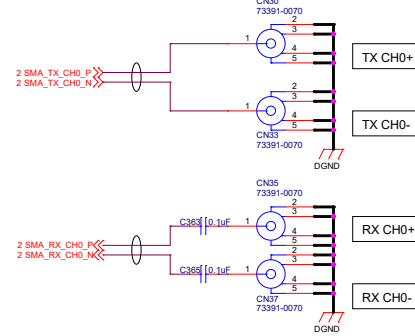
User Push Button Switch



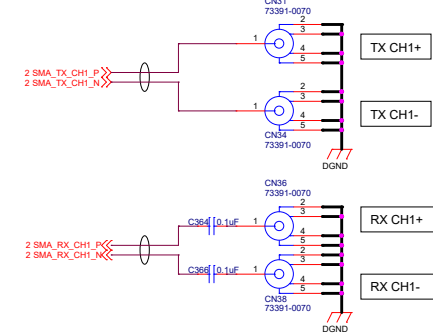
SMA XCVR REFCLK



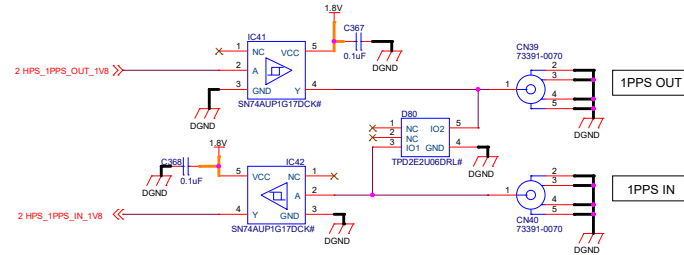
SMA TX/RX CH0

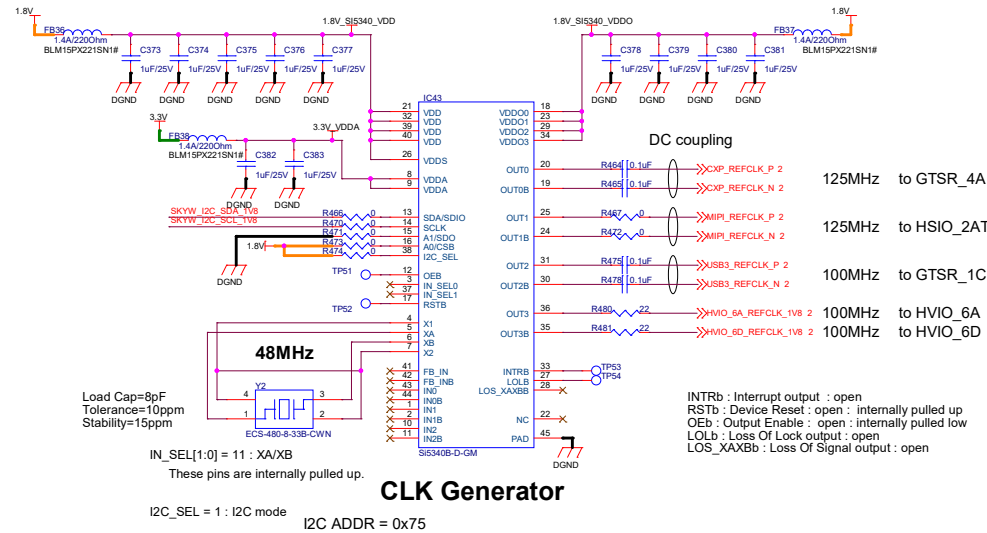


SMA TX/RX CH1

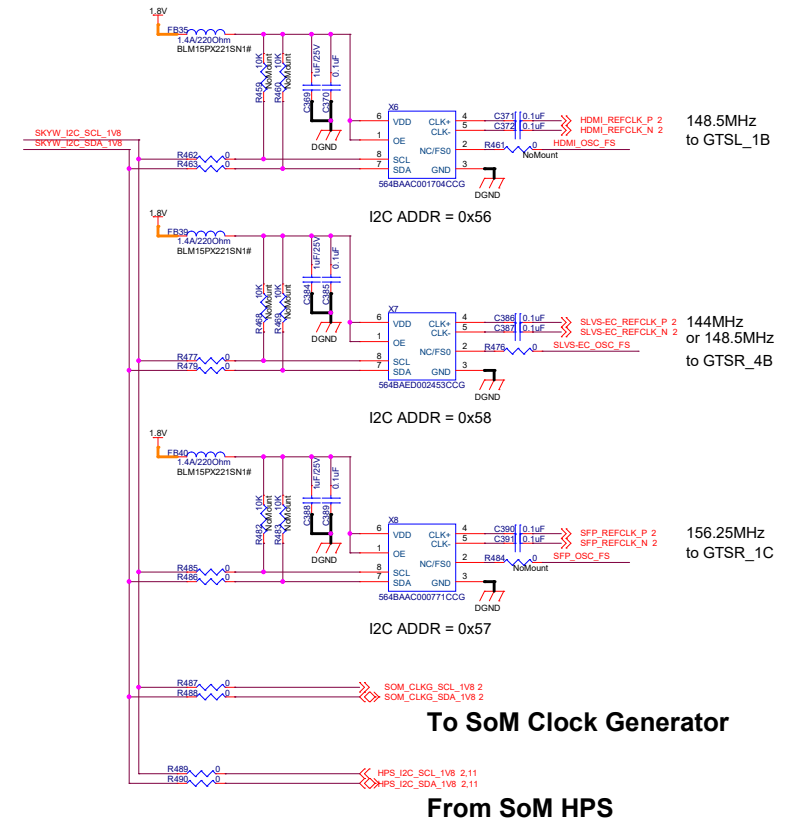
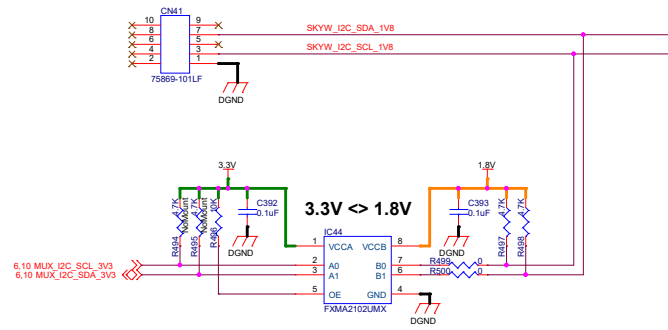


TSN 1PPS IN/OUT

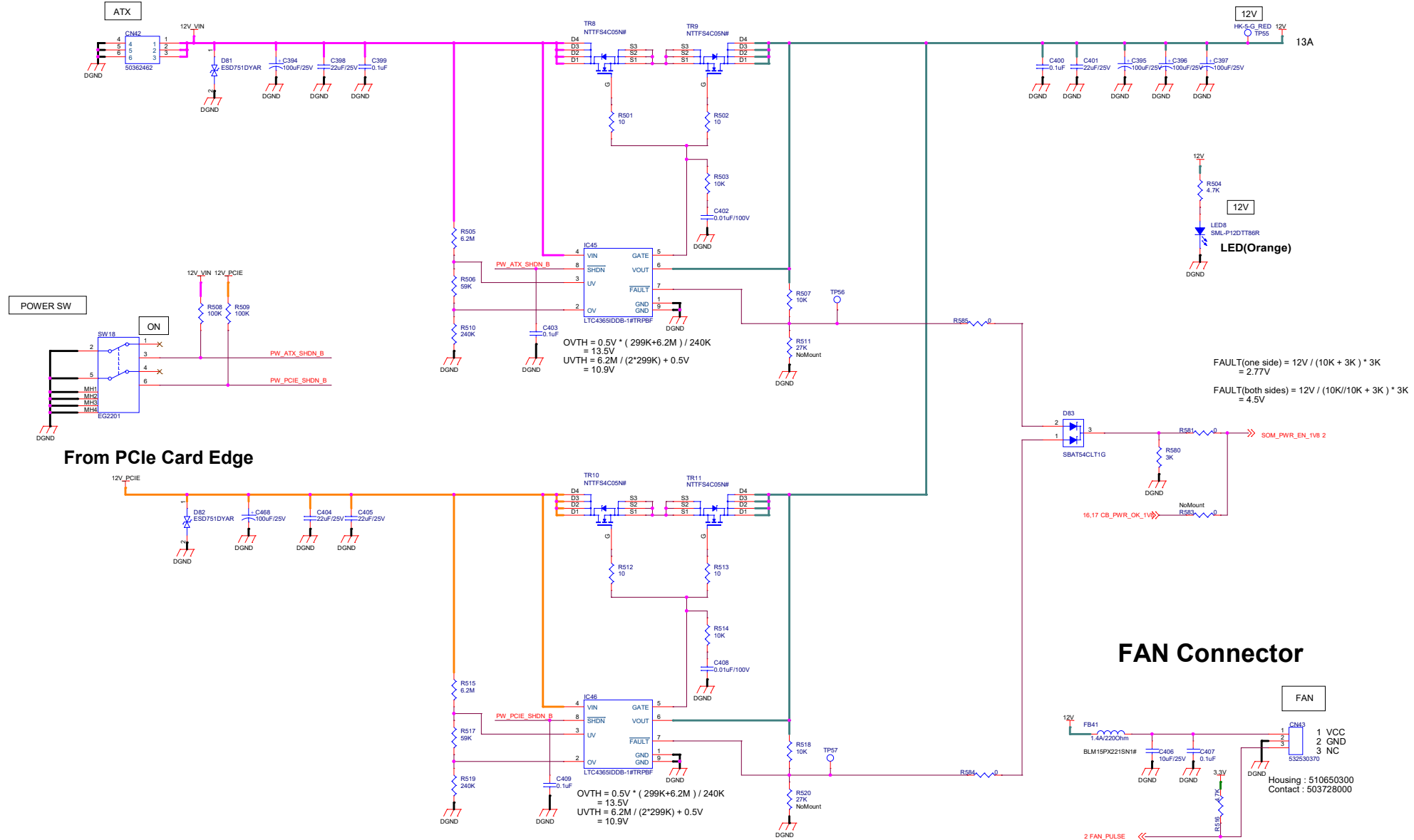




CBPROG-DONGLE connector



Power 12V



POWER 5V_SOM, 5V

5V_SOM

Vds=30V
Ids=60A

Irms=32A
Isat=50A

Vds=30V
Ids=60A

Vout = 0.6 * (1 + 11k / 1.5k)
= 5.0 V

tss = Ciss * Vf / Iss
= 0.22uF * 0.6V / 10 uA
= 13.2ms

Iout = 2*Rocset*Iocset+Voclos/Rdson + Ip-p/2
= 2*2.49k*9.5uA*-8mV/3.2m + 4.71A/2
= 14.64A

5V

Vout = 0.5 * (1 + 27k / 3k)
= 5.0V

EN (Vmax : 4V)
Vth (Rising) : 0.9Vmin
Vth (Falling) : 0.6Vmax

PGOOD (Vmax : 4V)
Vol : 0.4Vmax

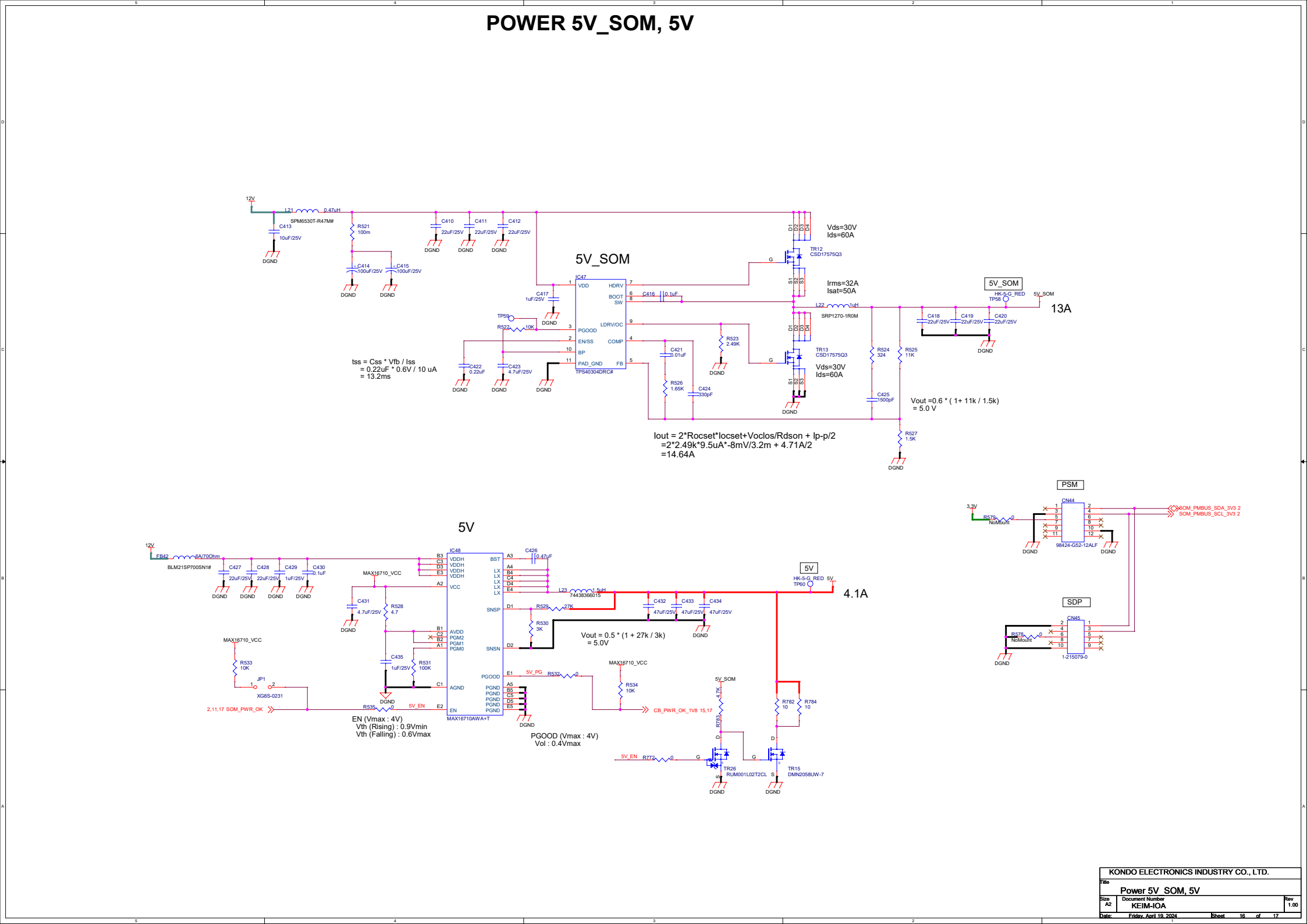
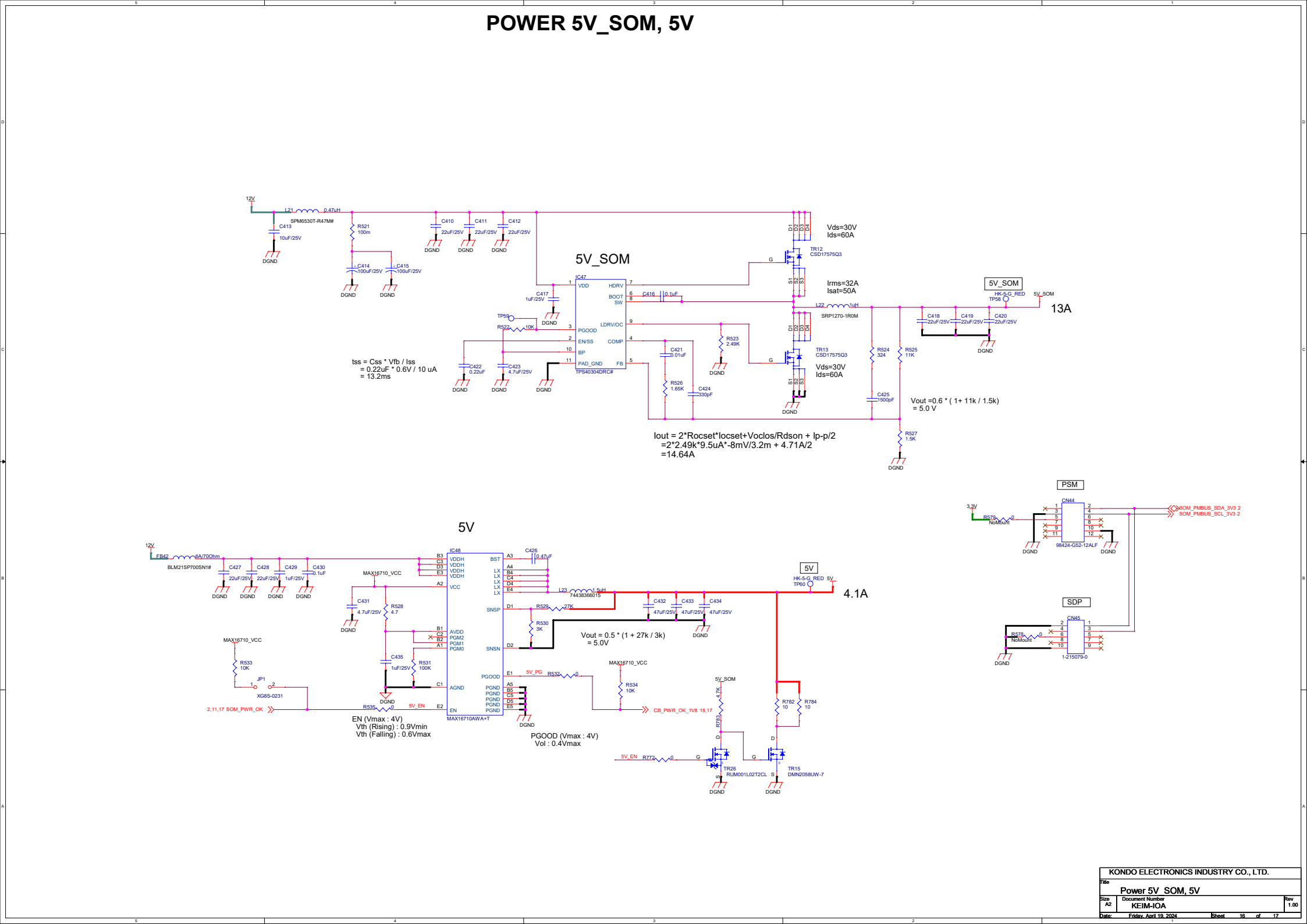
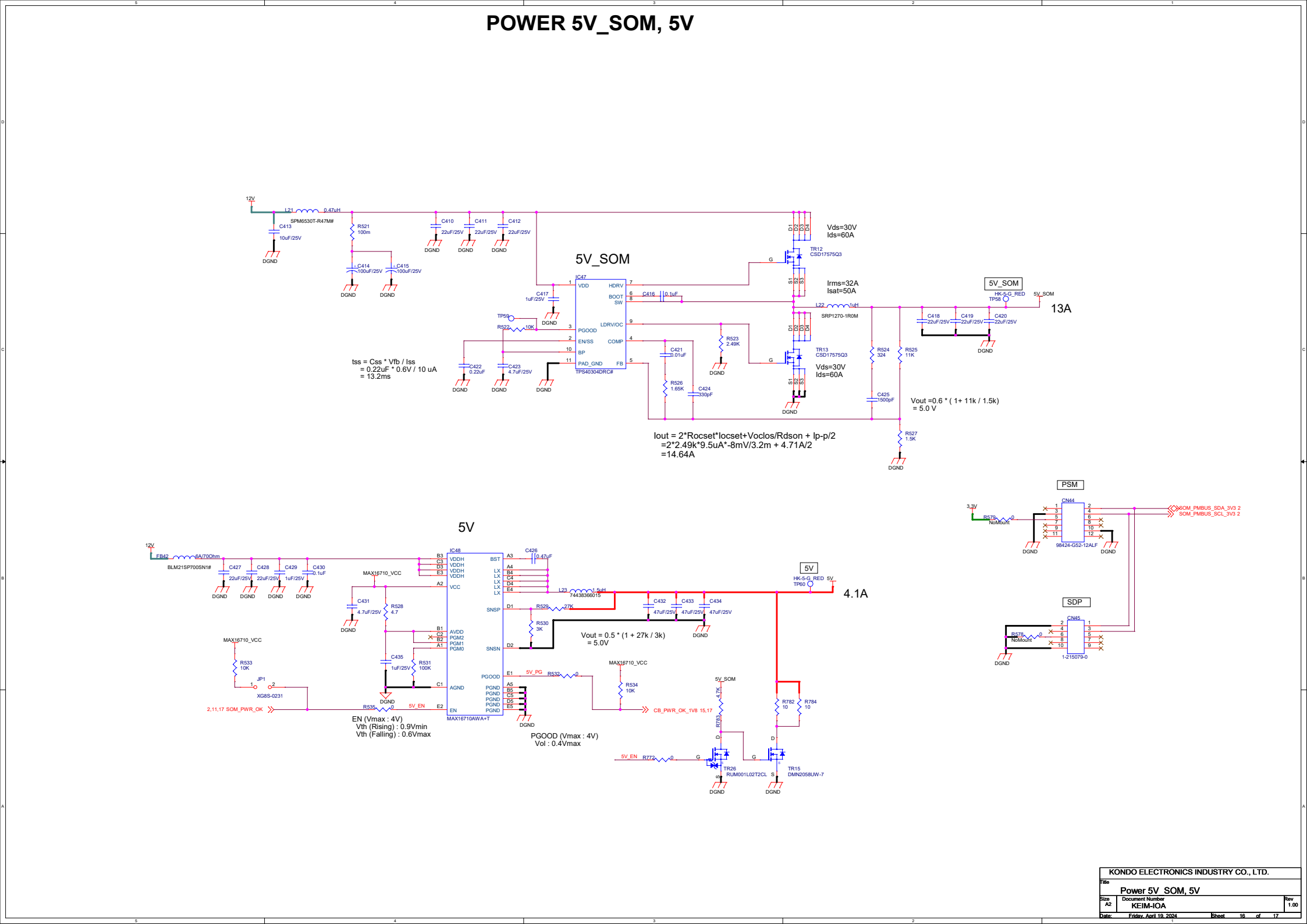
PSM

SOM_PMBUS_SDA_3V3 2
SOM_PMBUS_SCL_3V3 2

SDP

1-215079-0

KONDO ELECTRONICS INDUSTRY CO., LTD.			
File	Power 5V SOM, 5V		
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5V_SLVS-EC

